|INCH-POUND|

MIL-R-83726/27H(USAF)
9 August 1991
SUPERSEDING
MIL-R-83726/27G(USAF)
27 May 1988

MILITARY SPECIFICATION SHEET

RELAYS, HYBRID, TIME DELAY, VOLTAGE SENSITIVE, DELAY ON DROP-OUT, CLASS A, TYPE V, HERMETICALLY SEALED (POTTED), SPDT, 2 AMPERE, TRACK MOUNTED

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-83726.

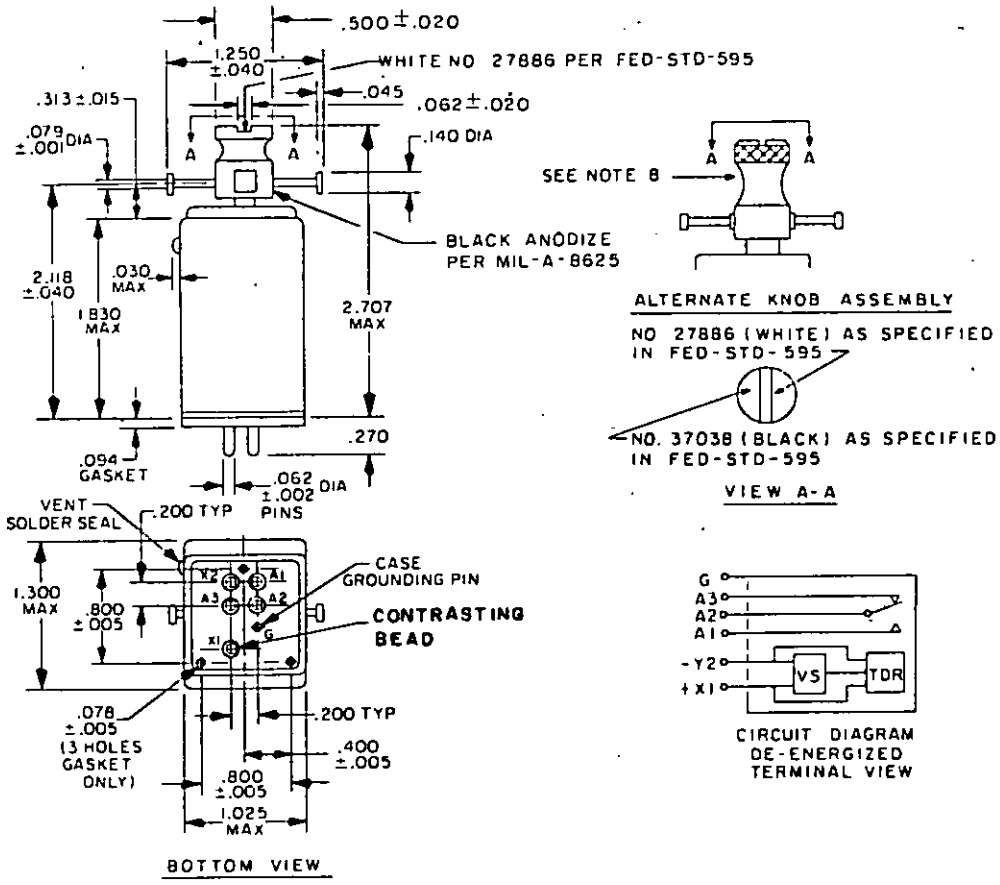


FIGURE 1. Outline dimensions and configuration of relay.

(H) denotes changes

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Inches	mm	Inches	mm
.001	0.03	.140	3.56
.002	0.05	.200	5.08
.005	0.13	.270	6.86
.015	0.25	.313	7.95
.020	0.51	.400	10.16
.030	0.76	.500	12.70
.040	1.02	.800	20.32
.045	1.14	1.025	26.04
.062	1.57	1.250	31.75
.078	1.98	1.300	33.02
.079	2.01	1.830	46.40
.094	2.39	2.118	53.80
		2.707	68.76

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Terminal numbers shall not appear on the relay header and there shall be a legible circuit diagram on relay which identifies each terminal location specified.
4. For relay mounting track see MIL-R-6106/3 and for relay bracket and socket assembly see MIL-R-6106/5.
5. Gasket material: Methylvinyl silicone (red). Shore hardness A40.
6. Terminal plating: Gold in accordance with MIL-G-45204, type II, class 1 over nickel plate.
7. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
8. Knob and post shall be nickel plated (100 microinches thick minimum) 303 stainless steel.

FIGURE 1. Outline dimensions and configuration of relay - Continued.

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REQUIREMENTS:

Operating data:

Configuration: SPDT.

Life/load ratings: See table I.

Timing action: See table II.

TABLE I. Life/load ratings (relay case grounded).

Type of load	Life (cycles)	Amperes 28 V dc
Resistive	*100,000*	*2.0*
Inductive	50,000	0.5
Lamp	100,000	0.2
Mechanical life (reduced current)	400,000	0.5
Intermediate current	100,000	0.50

Contact drop:

Initial: 0.150 volt maximum.

After life tests: 0.175 volt maximum.

Minimum current: 0.050 ampere.

Contact bounce: See table II.

Overload:

DC: 8 amperes minimum.

AC: Not applicable.

Rupture:

DC: 10 amperes minimum.

AC: Not applicable.

Input data:

Duty ratings: Continuous.

Input voltage:

Maximum: See table II.

Nominal: See table II.

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Voltage:

Operate: See table II.

Release: See table II.

Operate time: See table II.

Release time: See table II.

Maximum coil current: See table II.

TABLE II. Operating characteristics.

Dash number 1/	DC coil volts 2/			Maximum dc coil		Time-milliseconds-maximum		
	Nominal	Pickup 3/	Dropout	Volts	Amps	Operate 4/	Release delay 5/	Contact bounce 4/
-001	28	15.5 +0.5, -0.0	14.0 -0.0 6/	30	0.11	10	150 ±50	1.0
-002		25.75 ±0.25	24.75 ±0.25				3500 ±500	
-003		21.0 +0.5, -0.0	20.5 +0.0, -0.5					
-004		15.5 +0.5, -0.0	14.0 -0.0 6/				1000 ±200	

1/ The suffix letter W, X, or Y to designate quality level shall be added to the dash number.
2/ Over total operating temperature range.
3/ Applies also to MIL-R-6106 high temperature and continuous current tests.
4/ With nominal coil voltage.
5/ From dropout voltage to 8 volts. Below 8 volts, release delay time may be 0.
6/ Plus tolerance permits any value less than pickup voltage.

Electrical data:

Insulation resistance:

Initial: See table III.

After life: See table III.

Dielectric withstanding voltage (sea level):

Coil to case:

Initial: See table III.

After life: See table III.

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All other points:

Initial: See table III.

After life: See table III.

Dielectric withstanding voltage (80,000 feet):

Coil to case:

Initial: See table III.

After life: See table III.

All other points:

Initial: See table III.

After life: See table III.

Leakage current: See table III.

TABLE III. Electrical data.

Minimum insulation resistance-megohms		Dielectric withstanding voltage rms				
		Sea level 1/			80,000 feet altitude 2/	
Initial	After tests		Initial	After tests	Initial	After tests
100	100	Coil to case	1050	1050	350	350
		All other points	600	600	350	350
		Normal leakage current-rms microamperes 3/				
		Coil to case	25	25		
		All other points	25	25		

1/ Testing performed with relay installed in socket and bracket assembly (M6106/5-005) and secured in mounting track (M6106/3-001).
2/ Relay not installed in its socket and bracket assembly (M6106/5).
3/ Leakage current at sea level.

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Transients (applicable to -003 only): The time delay relay shall comply with MIL-STD-461 (applies to X1 and X2 terminals only; A1, A2, and A3 shall be wired but unloaded) and the requirements defined in method CE03, CS01, CS02, CS06, RE02, RS02, and RS03 of MIL-STD-462 except that the field intensity of RS03 shall be in accordance with table IV herein. Voltage spikes (transients of 500 microseconds or less duration) generated by the time delay relay and conducted on lines X1, X2 of the relay shall not exceed ±42 volts. Voltages generated by the time delay relays shall be monitored on the control/power leads (+X1, X2) with an oscilloscope having the ability to detect, measure, and display voltage spikes of 0.25 to 1,000 volts amplitude and 10 nanoseconds to 50 microseconds duration. The oscilloscope shall be capable of retaining the display. The qualification test report shall include photographs of all oscilloscope-displayed spike voltages. Switches used to control the time delay relays shall not be EMI-suppressed in any way. During susceptibility testing, the relay shall not change state nor have the timing interval affected in any way by the EMI signals. The timing interval shall be considered to include the instants of turn-on and turn-off.

TABLE IV. Field intensity levels.

Frequency	RS03 level (volts/meter)
14 kHz to 200 MHz	20
200 MHz to 450 MHz	65
450 MHz to 1 GHz	20
1 GHz to 12 GHz	100

Environmental data:

Temperature range (operating): -70°C to +85°C. +171°C nonoperating, 10 minutes. +200°C, nonoperating, 1 minute.

Altitude: 80,000 feet.

Shock:

G level: 50 g's.

Duration: 10 milliseconds.

Contact opening: 10 microseconds, maximum.

Contact closing: 1 microsecond, maximum.

Vibration (sinusoidal):

G level: 30 g's.

Frequency range: 10 Hz to 3,000 Hz.

Ⓗ Vibration (random): Applicable in accordance with MIL-STD-202, method 214, test condition 1B.

Power spector density: 0.4 g^2/Hz.

Frequency range: 50 Hz to 2,000 Hz.

Ⓗ Duration: 15 minutes each plane.

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Vibration scan:

Frequency range: 5 Hz to 2,000 Hz.

Intensity: 0.01 inch double amplitude or ±2 g's (whichever is less).

Acceleration: 15 g's.

Seal: Applicable to nonhermetically-sealed relays and hermetically-sealed (potted) relays.

Physical data:

Dimensions and configuration: See figure 1.

Weight: 0.24 pound (maximum).

Finish: Orange 32246, FED-STD-595.

Ⓗ Marking: See MIL-R-83726. In addition, relays shall be marked with the ESDS identifier as specified in MIL-STD-1285.

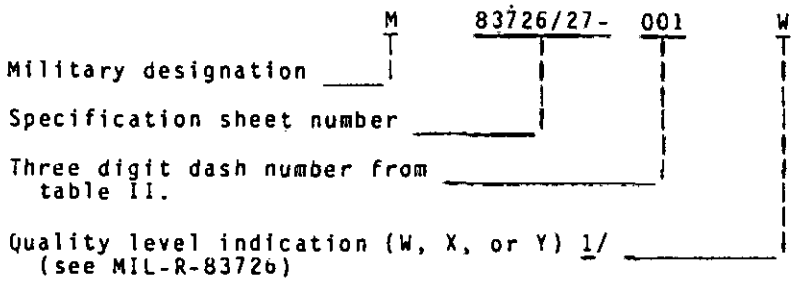
Ⓗ ESDS protection program: An ESDS protection program shall be implemented within 6 months of the date of revision H to this document. The manufacturer shall establish and maintain an ESD control program in accordance with MIL-STD-1686 for mission critical equipment. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification. This program shall be documented by an ESD control plan which must be under document control. As a minimum, this plan must address the identification of ESDS sub-components and end items, facilities, training, design protection, handling procedures, marking, cleaning, preservation, packaging, and quality assurance. A model ESD control program is available from the qualifying activity and may be used as a guideline. Further guidance for ESD control is available from the EOS/ESD Association and the Electronics Industry Association (EIA). This requirement is applicable to all manufacturers who handle ESDS component parts and materials in the relay manufacturing or testing process. This requirement is not limited to manufacturers qualifying ESDS end items.

Ⓗ ESDS verification: As a part of qualification or qualification after redesign, ESD testing shall be done in accordance with method 3015 of MIL-STD-883 modified to test at 16,000 volts. Testing at lower voltage levels is not required. This testing shall be accomplished as part of the group III for qualification inspection and as part of the group C inspection.

Ⓗ ESDS preservation and packaging: Relays shall be preserved and packaged in such a manner as to ensure that the integrity of ESD sensitive relays is not diminished. ESD sensitive relays shall be preserved and packaged in accordance with the requirements of MIL-STD-1686.

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Part or Identifying Number (PIN): Consists of the prefix M83726/27-, a three-digit dash number (expressed in milliseconds), and a quality level indicator as follows:



Example: M83726/27-001W

1/ Any relays numbered prior to the date of this specification without a quality level indicator shall be considered interchangeable (store and issue) with the "W" quality level.

CONCLUDING MATERIAL

Custodian:
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-F713-09)

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